PRANAV SHIVASHANKAR

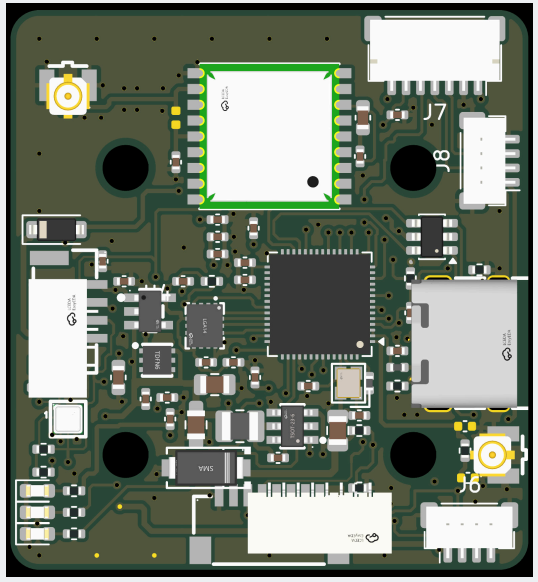
Engineering Portfolio · Summer 2027

B.S. Electrical & Computer Engineering · The University of Texas at Austin · Expected May 2029

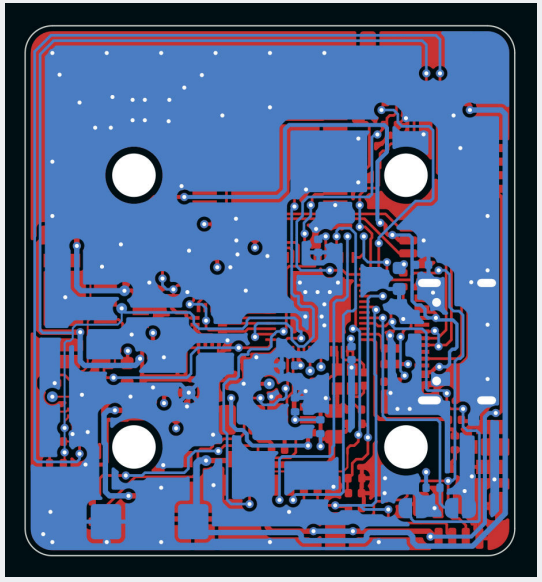
Austin, TX | 512-947-1154 | pranavshivashankar12@gmail.com | github.com/pranavs41

1 Swarm Drone Flight Controller: 4-Layer PCB

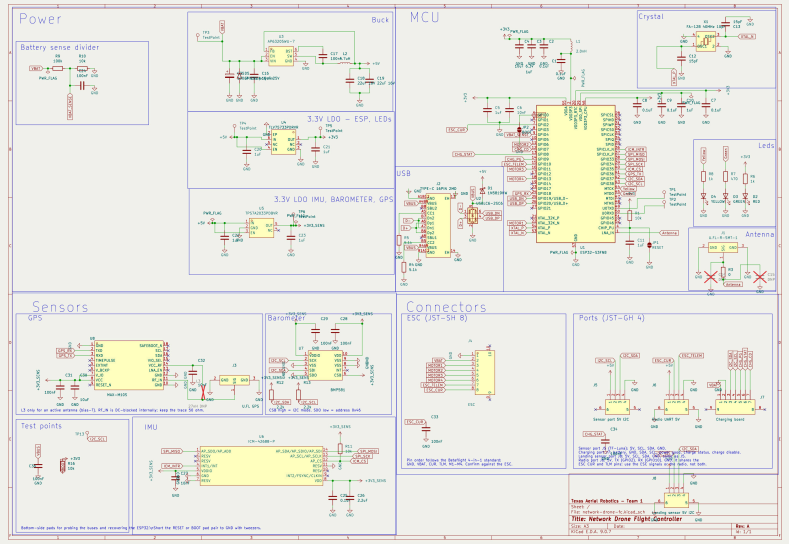
Texas Aerial Robotics · Lead Hardware Engineer · KiCad 9 · Sep 2026 to present



3D render, top side



Layout, outer copper layers



Schematic: power, MCU, sensors, connectors

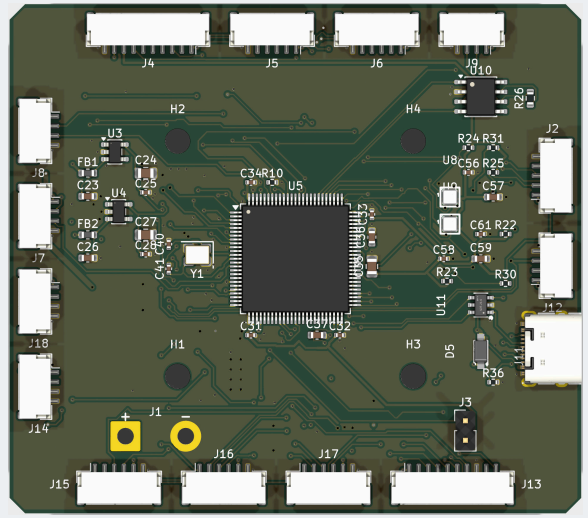
- 36 x 39 mm, 4-layer board built chip-down around a bare ESP32-S3 (QFN-56) with no modules, stacked on a 30 A 4-in-1 ESC
- Power tree from a 2S to 4S battery: a 2 A buck to 5 V, then separate 3.3 V LDOs for the MCU and a low-noise rail for the sensors
- ICM-42688-P IMU on SPI, BMP581 barometer on I2C, u-blox MAX-M10S GPS, and antenna feeds with pi matching footprints over solid ground
- Power budget modeled in Python: 359 mA peak on 3.3 V in a Wi-Fi transmit burst, 26 mV worst copper drop, every regulator under half its rating

4-layer	36 x 39 mm	0 / 0	359 mA
SIGNAL / GND / 3.3 V / SIGNAL	BOARD SIZE	ERC AND DRC ERRORS	PEAK 3.3 V LOAD

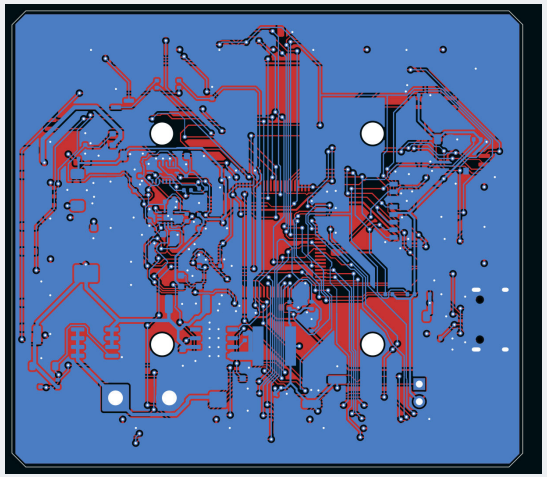
Network drones relay the radio link between the sensor drones and the ground station. Rev A passed six pre-order reviews; order files for OSH Park, OSH Stencils and DigiKey are ready.

2 VTOL Quadplane Flight Controller: 4-Layer PCB

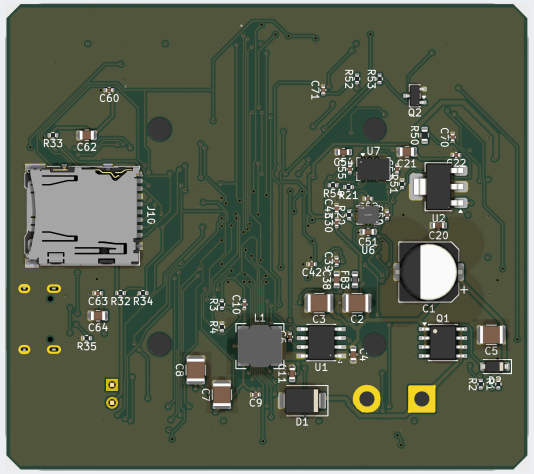
Personal project · STM32H743, ArduPilot pin map · KiCad 9 · Oct 2026



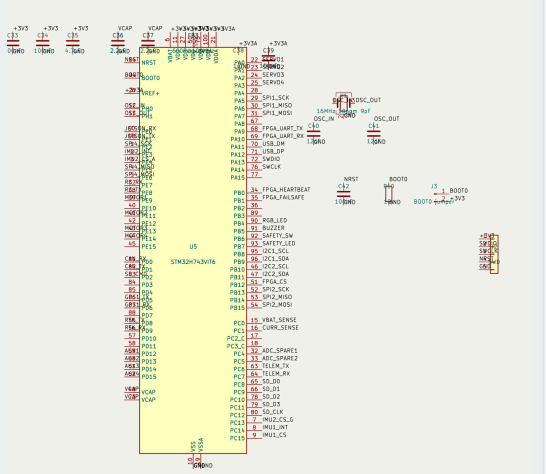
3D render, top side



Layout, outer copper layers



3D render, bottom side



Schematic: STM32H743 pin map

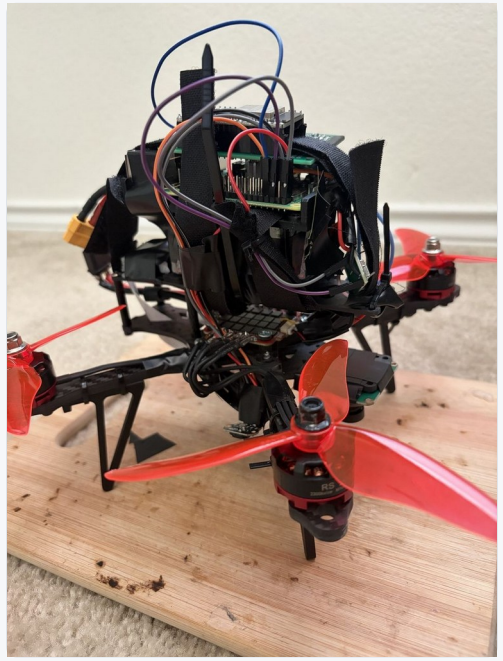
- 74 x 66 mm, 4-layer board around an STM32H743 (480 MHz): two dissimilar IMUs (ICM-42688-P, BMI088) on separate SPI buses, two BMP581 barometers, microSD logging
- 6S power path: reverse-polarity FET and TVS, a 3 A buck to 5 V, then three 3.3 V LDOs with a filtered rail per IMU so supply noise stays out of the gyros
- 14 JST-GH ports on Pixhawk pinouts (GPS, RTK, telemetry, CRSF, CAN, I2C, Jetson, FPGA link) and 12 PWM outputs on the 30.5 mm autopilot mounting pattern
- Schematic generated from Python blocks; KiCad run headless for scripted layout fixes, DRC, current and plane checks, and the JLCPCB fab package

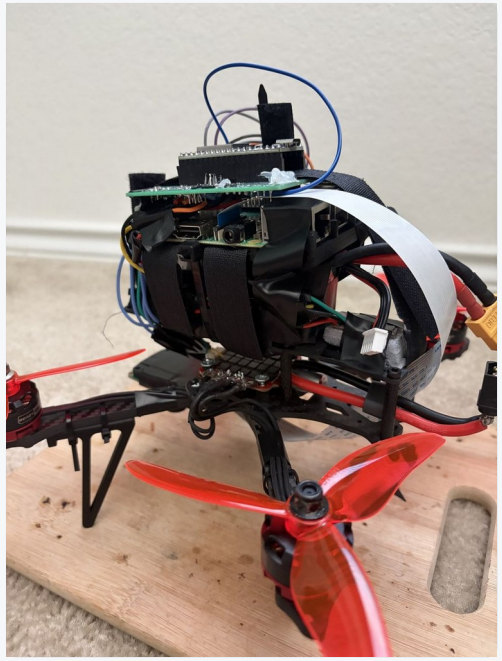
4-layer	74 x 66 mm	0	118
SIGNAL / GND / POWER / SIGNAL	BOARD SIZE	DRC ERRORS, 96 NETS	COMPONENTS, 338 VIAS

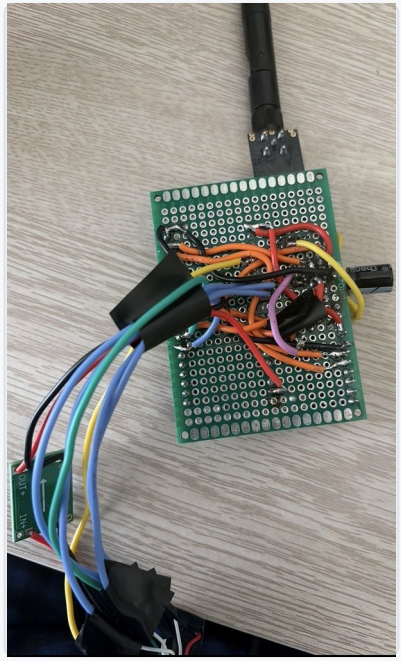
Designed for a 3.4 kg quadplane; fab package generated and quoted for assembly at JLCPCB. An FPGA coprocessor monitors it over the 10-pin link.

3 Custom FPV Quadcopter: Flight Controller

Designed, built, and flight-tested from scratch · ESP32 / C++ · Jun – Aug 2026







- 500 Hz cascaded control: outer angle loop (Kp 2.45) driving an inner rate loop (Kp 0.80, Ki 0.35, Kd 0.006), bounded to 15° attitude and 200°/s yaw
- BNO085 quaternion attitude at 200 Hz over I2C, clocked at 100 kHz because the sensor stretches the clock; DShot300 to four ESCs on GPIO 32/33/25/4 via the RMT peripheral
- Anti-windup: 0.85 decay on error sign crossing, 60 ms freeze window, integration gated above 60°/s gyro rate, 240 s leak, clamped at ±60 roll and ±40 pitch
- Failsafe ladder: 350 ms grace, link loss at 700 ms, descent at 120 units/s, touchdown below 0.08 m held 250 ms, then auto-disarm

500 Hz	200 Hz	DShot300	4
CONTROL LOOP	ATTITUDE UPDATE	ESC PROTOCOL	FAILSAFE STAGES

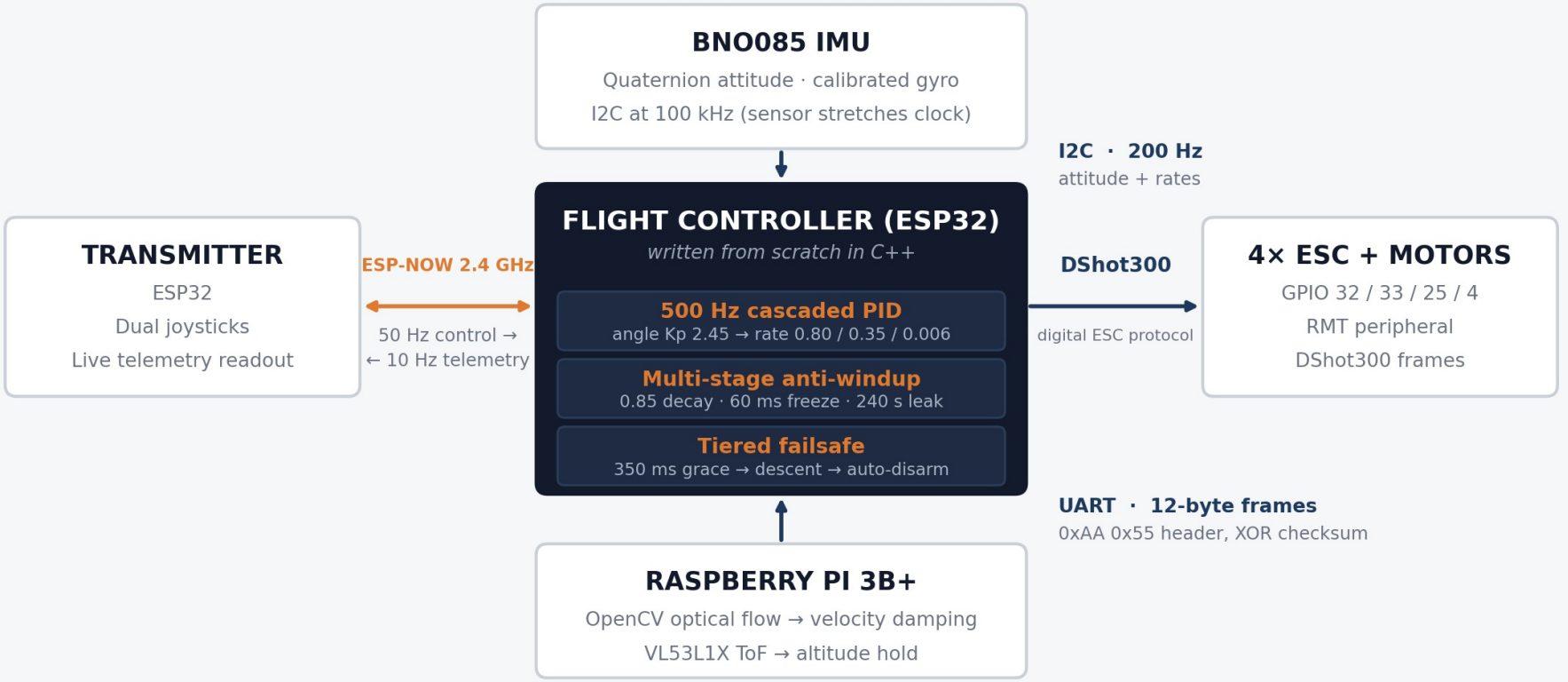
Firmware written entirely from scratch, with no flight-controller framework (Betaflight, ArduPilot, PX4).

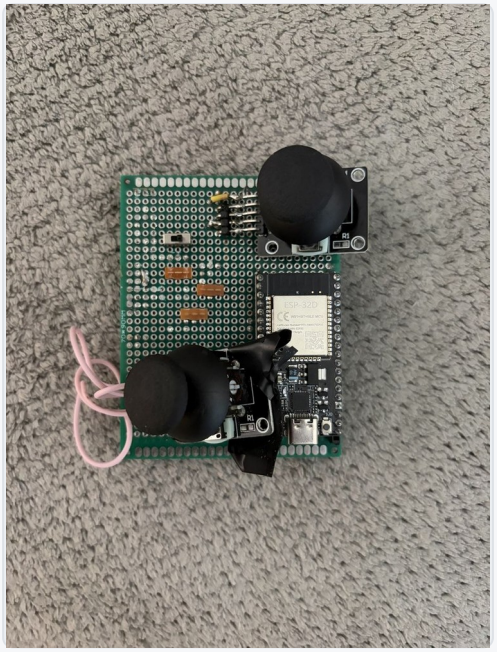
4 Quadcopter: Control and Autonomy Stack

System architecture, ground link, and companion-computer autonomy

Quadcopter System Architecture

Ground link, onboard control, and companion-computer autonomy





Dual-joystick transmitter

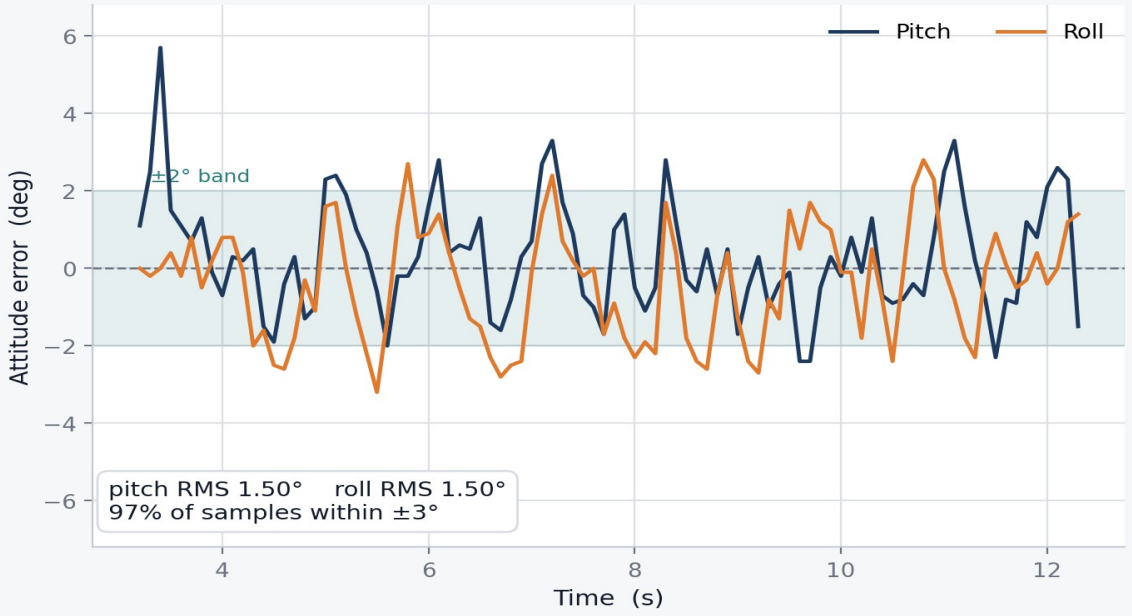
- Optical flow velocity damping: PD law at gain 0.40, engaging at flow quality 20 and dropping out at 12, authority capped at 3.5°
- VL53L1X altitude hold, PID 120 / 40 / 180, active between 0.40 m and 2.50 m with bumpless engagement
- Heading hold integrates yaw rate at Kp 1.5, limited to 60°/s and reset on stick input
- Trim solved from logged integrator steady state: PITCH_TRIM 6.0 to 8.2, ROLL_TRIM 0.95 to 0.7

5 Quadcopter: Flight Test Results

Measured from logged in-flight telemetry, not simulation

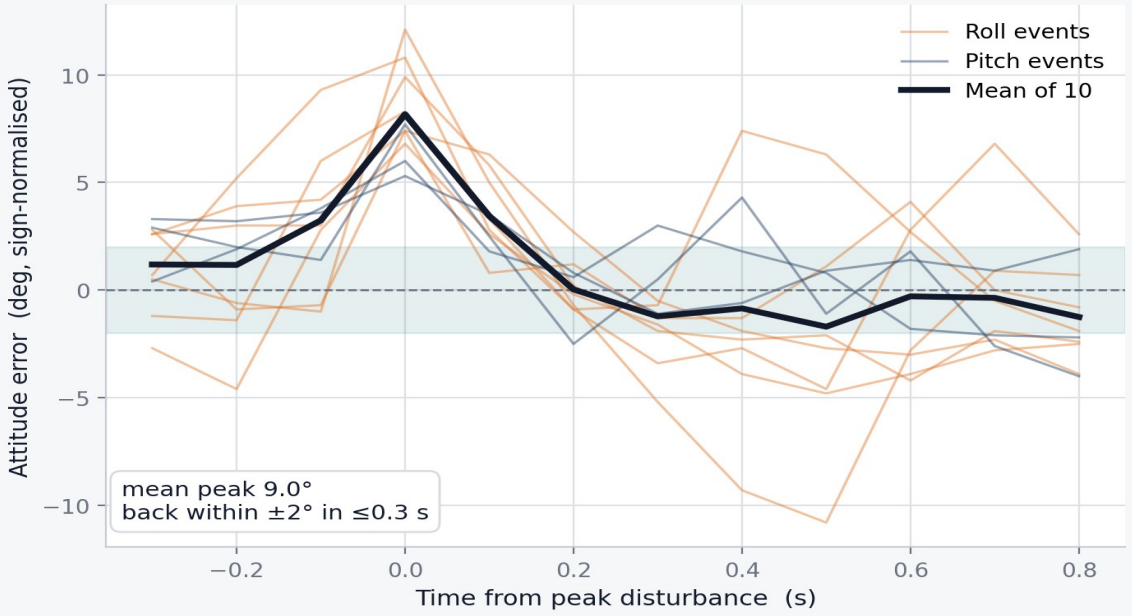
Hands-off hover: attitude hold accuracy

No pilot input on pitch or roll · 9 s of continuous hover



Disturbance recovery: 10 commanded tilts

Aircraft tilted on purpose, then released to self-level



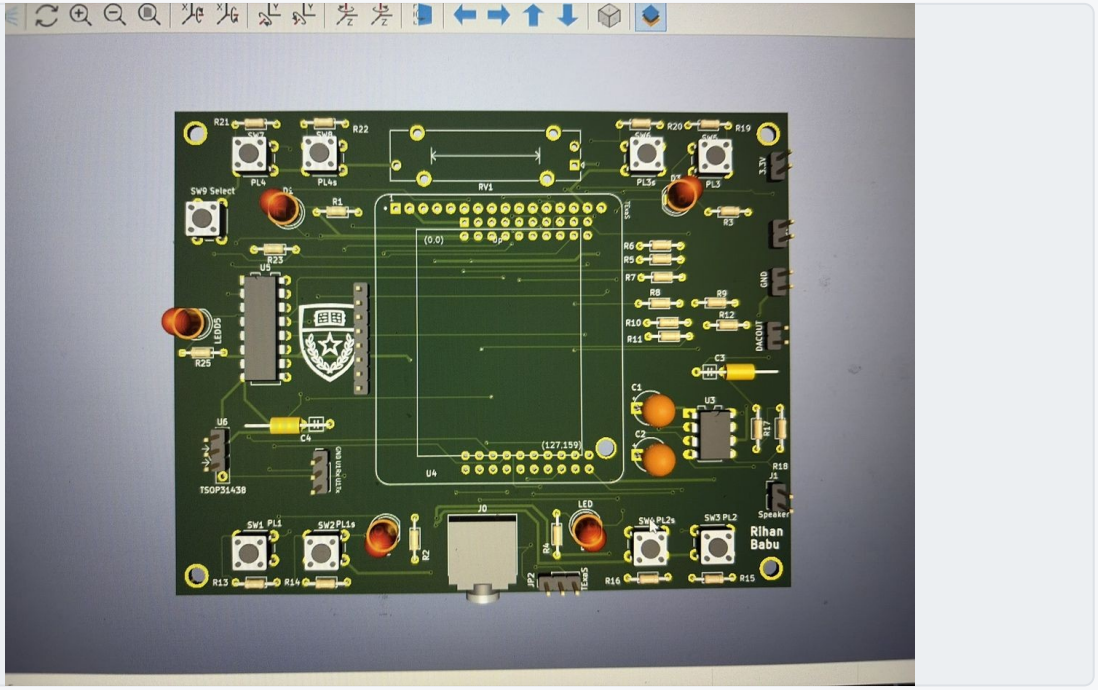
- Aircraft left to hold level with no pitch or roll input, measuring how tightly the controller keeps attitude in steady hover
- Aircraft tilted on purpose and released, measuring how quickly it drives the resulting error back to level

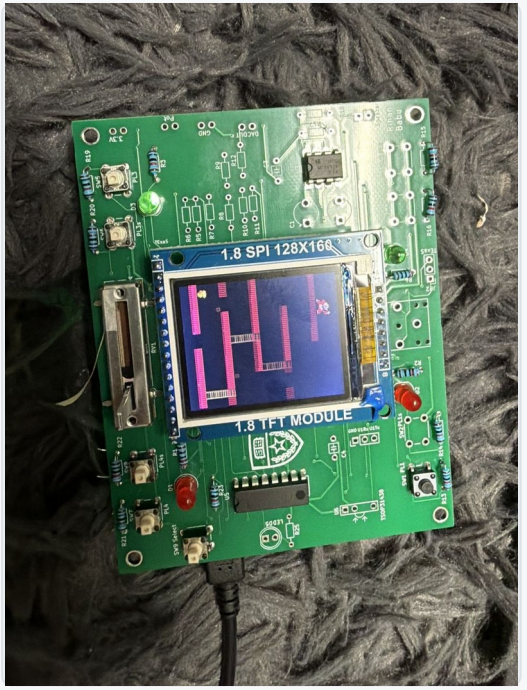
1.50°	97%	9.0°	≤0.3 s
HOVER RMS ERROR	WITHIN ±3°	MEAN DISTURBANCE	RECOVERY TO ±2°

Separate logged flights. Telemetry sampled at 10 Hz over the ESP-NOW downlink, so the 500 Hz control loop settles faster than the link can resolve.

6 Embedded Game Console: Custom PCB

TI MSPM0G3507 · ARM assembly and C · Mar – May 2026



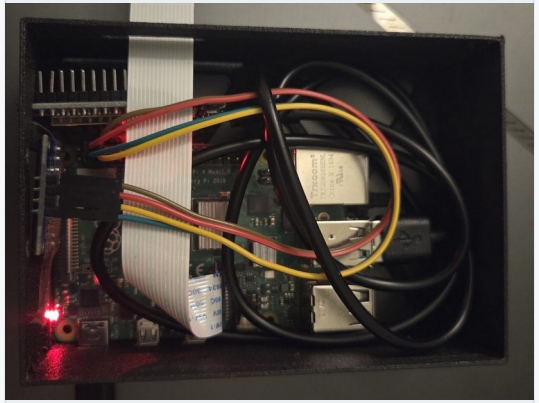


- Designed and assembled a custom PCB integrating buttons, LEDs, an LCD, a slide potentiometer, and audio output
- Wrote a Donkey Kong-style infinite platformer in ARM assembly and C
- Implemented LCD graphics rendering, a title screen, language selection, and sound effects
- Full stack from schematic capture through board bring-up to shipped firmware

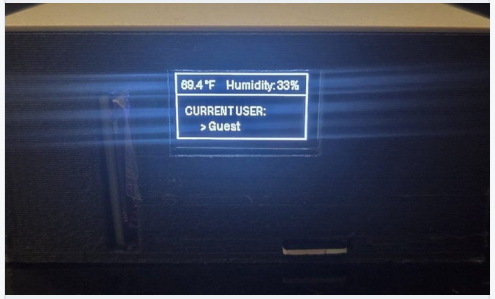
7 FridgeGuard: 3rd Place, UT IEEE Techathon

Raspberry Pi 4B + ESP32 · computer vision and BLE attribution · Apr 2026

- Shared-living-space fridge security system built and presented in a single hackathon weekend
- Captures before/after frames via 500 ms sampling between door-open and door-close events
- Groq Llama 4 Scout vision API compares image pairs against a SQLite inventory database to identify the specific item added or removed, then updates stock state
- ESP32 scans roommates' phone BLE advertisements to attribute each fridge action to a specific user



Pi 4B + wiring



OLED status display

Detection pipeline

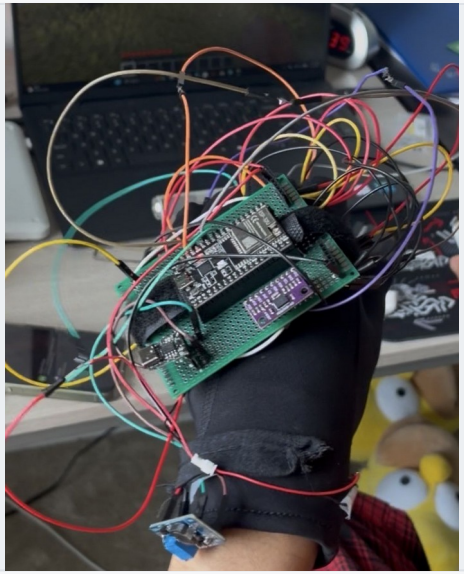
1. Door-open event detected
2. Frames sampled every 500 ms until close
3. Before/after pair sent to Llama 4 Scout
4. Diff resolved against SQLite inventory
5. BLE scan attributes action to a user

3rd	500 ms	2	48 hrs
PLACE, UT IEEE	FRAME SAMPLING	MCUs INTEGRATED	BUILD TIME

8 Gesture-Control Glove: Two Firmware Targets

ESP32 · BNO085 IMU, hall-effect and contact sensing · Feb – Apr 2026





- Four fingertip contact pads on GPIO 13/14/27/26 plus two hall-effect sensors that register a closed fist when index and ring trigger together
- BNO085 quaternion attitude at 50 Hz, zeroed against a measured neutral hand pose and guarded by a five-attempt init retry so a missing IMU disables tilt instead of hanging
- Sensor frame rotated into the control frame by summing and differencing the pitch and roll deltas, so a naturally angled hand maps cleanly to two orthogonal axes

Drone mode · ESP-NOW transmitter

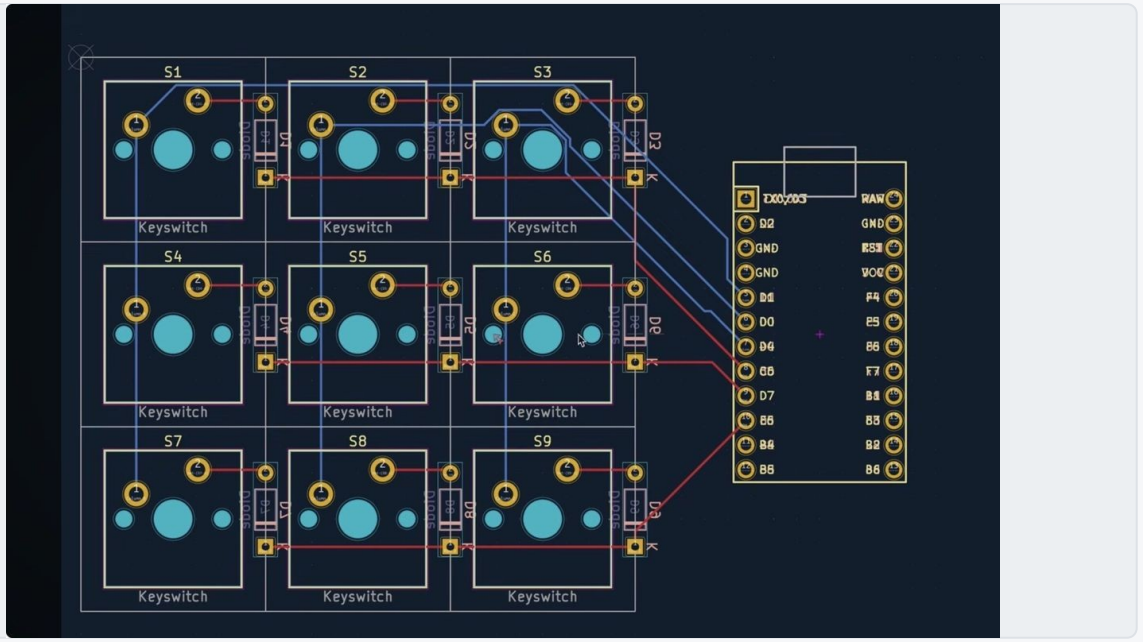
- Packed 10-byte struct byte-matched to the transmitter: pitch, roll and yaw scaled to ±1000 with a 13° deadzone and 65° full deflection, sent at 50 Hz
- Ring finger acts as a modifier that reroutes lateral tilt from roll to yaw; index commands throttle up, middle commands land
- Redundant kill on pinky or fist, while arming stays on the transmitter's physical switch by design and is never exposed to the glove

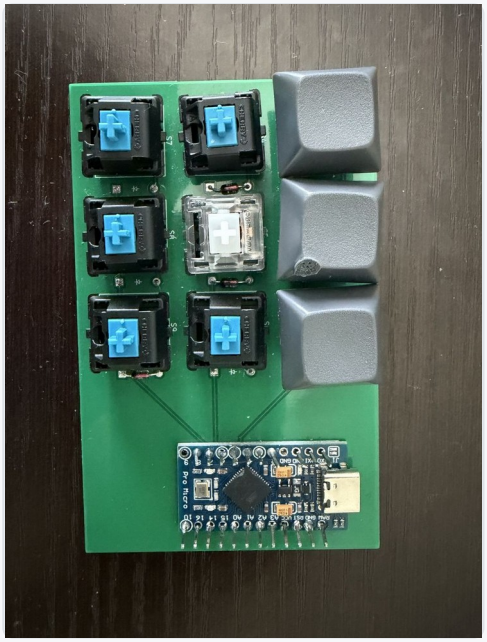
PC mode · BLE HID keyboard

- Same hardware pairs to Windows as a standard keyboard with no host-side drivers, using edge-detected press and release for true key-hold behaviour
- Tilt drives strafe and walk past a 40° threshold; index holds jump, middle latches sprint as a toggle, ring and pinky tap inventory and sneak
- Closed fist sends escape behind a one-second debounce

9 Custom Macropad: Keyboard PCB Design

KiCad · Arduino Pro Micro (ATmega32U4) · QMK firmware





- Designed a 3×3 macropad in KiCad from schematic capture through PCB layout and fabrication
- Laid out a switch matrix with per-key 1N4148 diodes to prevent ghosting and enable reliable n-key rollover
- Hand-soldered Cherry MX switches, diodes, and an Arduino Pro Micro (ATmega32U4) controller
- Flashed QMK firmware with a custom keymap, taking the board from bare PCB to a working USB HID device

Full source and design files for every project: github.com/pranavs41 · pranavshivashankar12@gmail.com